

DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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Subject: Python Programming

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Lab Assignment: 20

Aim:

To write a Python program to generate an Abecedarian series from a list of words.

Objective:

- To understand the concept of Abecedarian words.
- To implement a Python function to check whether letters in a word appear in alphabetical order.
- To display all Abecedarian words from user input.

Theory:

An **Abecedarian word** is a word in which the letters appear in **alphabetical (dictionary) order**. The letters may repeat, but they should never appear in decreasing alphabetical order.

Example:

- **"almost"** → Abecedarian (a < l < m < o < s < t)
- **"hello"** → Not Abecedarian (since l comes after e, but e appears again later)

In Python, we can check this by:

1. Converting the word into a list of characters.
2. Sorting the characters alphabetically using `sorted()`.
3. Comparing the original list with the sorted list.
If both are the same, the word is Abecedarian.

Algorithm:

1. Start the program.
2. Define a function `is_abecedarian(word)`.
3. Convert the word into a list of characters.
4. Sort the list using `sorted()`.
5. Compare the original list with the sorted list.
6. If both are equal, return **True**, otherwise return **False**.
7. Accept words from the user separated by space.
8. Split the input into a list of words.
9. For each word, call the function `is_abecedarian()`.
10. If the function returns **True**, print the word.

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11. Stop the program.

Pseudo Code:

START

```
DEFINE function is_abecedarian(word)
    IF list(word) == sorted(word)
        RETURN True
    ELSE
        RETURN False
```

```
INPUT words separated by space
SPLIT input into list of words
```

```
PRINT "Abecedarian words are:"
```

```
FOR each word in list
    IF is_abecedarian(word)
        PRINT word
```

END

Code:

```
def is_abecedarian(word):
    return list(word) == sorted(word)

words = input("Enter words separated by space: ").split()

print("Abecedarian words are:")
for w in words:
    if is_abecedarian(w):
        print(w)
```

Output:

```
Enter words separated by space:  heap almost cat effort
Abecedarian words are:
almost
effort
```

Conclusion:

Thus, the Python program was successfully implemented to identify and display Abecedarian words, where the letters in each word appear in alphabetical order.